Hybrid guessing of 1–2 unknowns in the six-point Riemann–Roch / 2-adic chain system

2026-10-15 (UTC) · follow-up to `REPORT.md` / `report.pdf` · experiment log (observations; no hypothesis status changed). Full log: `HYBRID_REPORT.md`; table `hybrid_runs.csv`; data `data/hybrid1/`.

Setup

Circuit encoding, `mult` arm ($p = 268496897$, $S = \{X : X^{2^k} = c\}$), seed 1. We fix g of the five unknowns: $g = 1$ fixes c_1; $g = 2$ fixes c_1, c_0 (so $C(x)$ is fully known). Guessed variables are substituted out. Three branches per (k, g): **planted/true** (the guess equals the planted witness; the consistent branch), **planted/wrong**, and **random/wrong** (random T). Solver: msolve 0.10.1 F4, grevlex, 1 job, 1800 s limit; 15 GiB host. Code at commit 8ceaf173b3; run started 20261015T1626Z. A cell that does not complete at some k is skipped for larger k. The $g = 0$ rows are from the earlier sweep.

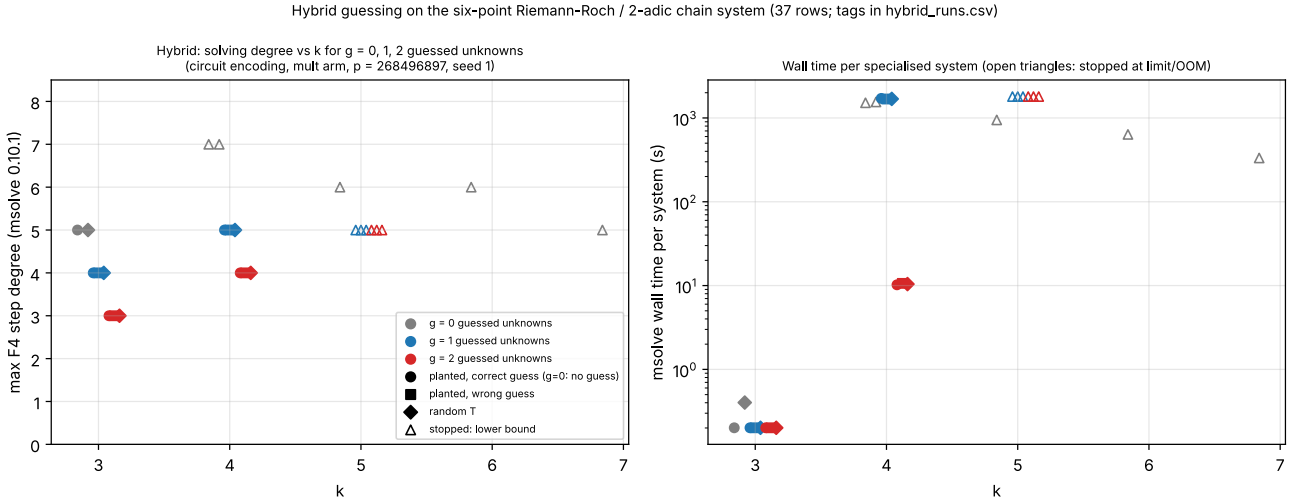


Figure 1: Left: max F4 degree vs k for $g = 0, 1, 2$ (filled = complete; open triangle = lower bound from timeout or OOM). Right: wall time per specialised system. Run tags in `hybrid_runs.csv`.

Results (measured)

k	g	n	stop	max F4 deg	wall / system	peak RSS
3	0	14	complete	**5**	0.2–0.4 s	13 MB
3	1	13	complete	**4**	0.2 s	13 MB
3	2	12	complete	**3**	0.2 s	13 MB
4	0	25	OOM	≥ 7	≈ 1530 s	13.6 GB
4	1	24	complete	**5**	1683–1711 s	2.2 GB
4	2	23	complete	**4**	10.2–10.6 s	94 MB
5	0	36	OOM	≥ 6	947 s	13.6 GB
5	1	35	timeout	≥ 5	1800 s	2.5 GB
5	2	34	timeout	≥ 5	1800 s	5.1 GB

Controls: every complete planted/true system returned exactly 1 solution; every complete planted/wrong and random/wrong system returned no solution (`[-1]`). Within each (k, g) the three branches have the same max F4 degree and wall times within 2%, so rejecting a wrong guess costs as much as solving the right one. At $k = 5$ the runs were inside a degree-5 F4 step when they stopped ($g = 2$: 267807×285940; $g = 1$: 164866×246582).

Derived

- Solving degree with $g = 2$: 3, 4, ≥ 5 for $k = 3, 4, 5$. With $g = 1$: 4, 5, ≥ 5. With $g = 0$: 5, ≥ 7. Each guessed unknown lowers the degree by 1 at $k = 3, 4$.
- At $k = 4$ the second guess cut the wall time by a factor of about 165 (≈ 1700 s ⟶ 10.4 s).
- A guess is an element of F_p. Exhaustive guessing multiplies the number of systems by p^g (2^{224g} for P-224), against a per-relation budget of $\approx 2^{27.8}$. Equivalently, a specialised family has an expected $|S|^6 / (720p) \cdot p^{-g} \approx 2^{-9.5-224g}$ relations at the $N^{1/3}$ parameters.

Inference, limitations, next checks

Inference: guessing lowers the degree by a constant per guessed unknown but does not stop its growth in k (with $g = 2$: 3 ⟶ 4 ⟶ ≥ 5). Guessing over F_p cannot pay for itself in the relation search, because each guess divides the expected relation count

by p. The degree evidence rests on complete runs at $k \le 4$ and lower bounds at $k = 5$; it is consistent with growth in k but not a law. **Limitations:** one seed and one arm; only the coefficients of C were guessed; timeouts are operational facts; 15 GiB host. **Next:** guess the remainder or quotient variables of the last squaring steps; rerun $k = 5$, $g = 2$ with a longer limit; restrict guesses to structured subsets, which needs its own relation-count model.